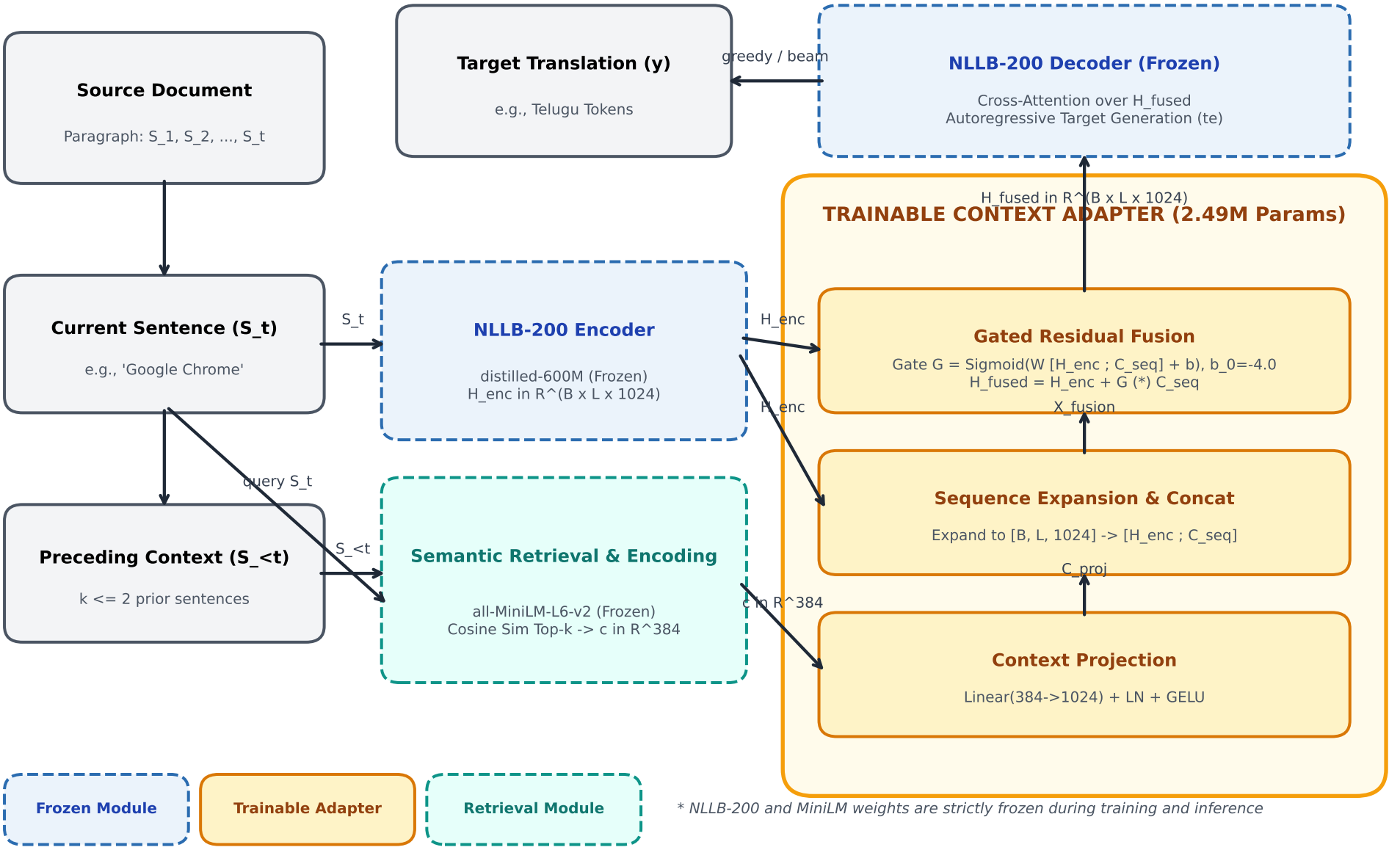
Source Document
Paragraph: S_1, S_2, ..., S_t
Current Sentence (S_t)
e.g., 'Google Chrome'
Preceding Context (S_<t)
k <= 2 prior sentences
S_t
query S_t
S_<t
NLLB-200 Encoder
distilled-600M (Frozen)
H_enc in R^(B x L x 1024)
Semantic Retrieval & Encoding
all-MiniLM-L6-v2 (Frozen)
Cosine Sim Top-k -> c in R^384
H_enc
H_enc
c in R^384
TRAINABLE CONTEXT ADAPTER (2.49M Params)
Gated Residual Fusion
Gate G = Sigmoid(W [H_enc ; C_seq] + b), b_0=-4.0
H_fused = H_enc + G (*) C_seq
X_fusion
Sequence Expansion & Concat
Expand to [B, L, 1024] -> [H_enc ; C_seq]
C_proj
Context Projection
Linear(384->1024) + LN + GELU
H_fused in R^(B x L x 1024)
NLLB-200 Decoder (Frozen)
Cross-Attention over H_fused
Autoregressive Target Generation (te)
greedy / beam
Target Translation (y)
e.g., Telugu Tokens
Frozen Module
Trainable Adapter
Retrieval Module
* NLLB-200 and MiniLM weights are strictly frozen during training and inference